

LEONARDO RABELLO MANZO

AI SOLUTIONS & AUTOMATION ENGINEER | CONTECH | PYTHON, LLMS & APIS

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PROFESSIONAL SUMMARY

AI Solutions and Automation Engineer with a Postgraduate Degree in AI Engineering and over 10 years of experience in construction operations, project management, and process improvement. Builds applied solutions in Python and generative AI using REST APIs, OAuth, SQL/SQLite, document search, multimodal AI, and structured outputs. Combines domain expertise, operational problem analysis, and technical execution to translate business needs into automations, assistants, and integrations—aligned with AI Solutions, AI Automation, Implementation, ConTech, and PropTech roles.

TECHNICAL SKILLS

- **AI & LLMs:** LLM APIs, LangChain, structured outputs, prompt engineering, multimodal APIs, document search, SQLite FTS5.
- **Development & Integration:** Python, SQL, SQLite, JavaScript, JSON, NDJSON, REST APIs, OAuth, Git, GitHub, unit testing.
- **Data & Applications:** Streamlit, Plotly, OpenCV, local desktop apps, data persistence, secure token storage.
- **Implementation & Domain:** Requirements analysis, process improvement, project management, stakeholders, construction operations, ConTech / PropTech.

SELECTED AI & AUTOMATION PROJECTS

Síndico Inteligente Python, SQLite FTS5, PDFs, JavaScript

- Local desktop app for condominium management, featuring PDF import/indexing, full-text search citing document and page numbers, reservation rules, history, SQLite backup, and JSON export.
- Implemented preview before mutations and conflict locking; automated suite validated with 4 passing tests.

Agenda Retrátil Python, Google Calendar API, OAuth, SQLite, Keyring

- Desktop assistant for reading, creating, editing, and deleting events only after preview and explicit confirmation.
- Implemented secure token storage in Windows Credential Manager, local persistence, replay protection via request_id, and overwrite protection via etag; 4 passing automated tests.

Physical-Financial Schedule Generator Python, Streamlit, LLM API, Plotly

- Transforms free-text construction scope into structured JSON, calculates start/end dates and dependencies, and renders stages, budget, and Gantt chart with error handling.

Vigia Obra Safety Python, OpenCV, Multimodal API, NDJSON

- Computer vision assistive pipeline sampling video frames and flagging safety non-conformities into auditable NDJSON logs for human review.

PROFESSIONAL EXPERIENCE

Arquitetool Aug/2026 – Present

AI & Automation Consultant (temporary) • São Paulo, Brazil

- Works at the intersection of construction, processes, product, and technology, supporting the identification and evolution of digital solutions applied to the sector.
- Translates real job-site and operational needs into product and automation opportunities, bridging 10+ years of construction management experience with scalable tech solutions.
- Fosters alignment between business context, operations, and tech, contributing to user-centered digital decisions for construction.

CH3 Incorporadora Oct/2025 – Present

Construction Manager • São Paulo, Brazil

- Conceived and developed a construction checklist app to standardize field logs and support operational control; financial impact pending measurement.
- Manages internal/subcontractor teams, executive scheduling, physical-financial tracking, quality, procurement, and vendor/designer interface.

Blink Reformei Jun/2024 – Oct/2025

Construction Coordinator • São Paulo, Brazil

- Coordinated project schedules, purchasing, field technical compatibility, contracts, reporting, and client/vendor communication.

Torão Design Mar/2022 – 2024

Construction Coordinator • São Paulo, Brazil

- Managed planning, costs, quality, KPIs, and change order management for commercial and residential construction.

Engconsult 2023

Construction Manager (temporary) • São Paulo, Brazil

- Schedule, procurement, progress measurement, risk, and technical interfaces for retail store (Caedu chain) and shopping mall projects.

M. Mazza Arquitetura Mar/2021 – Mar/2022

Construction Coordinator • São Paulo, Brazil

- Coordinated legal permits, purchasing, logistics, technical compatibility, and site inspection.

Arq. LManzo Jan/2016 – Mar/2021

Independent Architect • São Paulo, Brazil

- Developed residential and commercial architectural designs, permit approvals, specifications, quantities, and site supervision.

Prior Professional Background

2008 – 2015

Bougue (Architectural Designer 2014-2015); Torão Design (Designer Mar/2012-2014); Internships: Construtora Bracoo (2011-2013), União Técnica (2009-2011), Nucleora (2008), Monaco San Marzano (2008).

EDUCATION

- **Postgraduate Degree in AI Engineering:** Faculdade Impacta Digital • Completed 2026
- **Postgraduate Degree in Project Management:** Universidade Anhembi Morumbi • 2023
- **B.A. in Architecture & Urbanism:** Centro Universitário Belas Artes de São Paulo • 2007 – 2013

CERTIFICATIONS & COMPLEMENTARY COURSES

Certifications: Google AI Professional Certificate (Google / Coursera); Claude Platform 101 (Anthropic); Gesture Recognition System with Python & ML (DIO).

Courses: Lean Construction (Udemy); Scrum Methodology (Udemy); Finance for Small Business (Sebrae); Residential Electricity & NR 10 (Seal).

BUSINESS, DOMAIN & PROFESSIONAL TOOLS

Business & Domain: Project Management, Requirements Analysis, Planning & KPIs, Process Improvement, Risk Management, Multidisciplinary Leadership, Technical-Business Communication, Technical Documentation, Construction Operations, ConTech / PropTech.

Tools: Excel, Power BI, MS Project, Office Suite, AutoCAD, SketchUp/Layout, Lumion, V-Ray, Twinmotion, Photoshop.

LANGUAGES & ADDITIONAL INFORMATION

Languages: Portuguese (Native), English (Advanced). • **Availability:** Driver's License Category B • Travel Availability • Preference for remote work, open to hybrid or onsite roles in São Paulo & Osasco.